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ECG Arrhythmia Detection and Classification using Deep Learning on the MIT-BIH Arrhythmia Database

**Synopsis report on
Deep Learning Project
[ICT-4442]**

Submitted By

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SYNOPSIS

Team Members

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Title of the Project: ECG Arrhythmia Detection and Classification using Deep Learning on the MIT-BIH Arrhythmia Database

Brief Description of the Project

Arrhythmias can be difficult to detect because they may occur only briefly during long ECG recordings. We will compare four models for heartbeats classification based on the MIT-BIH Arrhythmia Database and five AAMI classes (N, S, V, F, Q). ECG recordings will be retrieved from PhysioNet via the wfdb library, and heartbeats will be extracted around annotated R-peaks.

We will compare four classifiers: a Random Forest using wavelet and statistical features, a 1D CNN using raw ECG signals, a BiGRU, and a Transformer encoder. This allows us to compare a traditional ML model with three deep-learning architectures. As there are about 83% of normal heartbeats in the dataset, the accuracy is not always a good measure. That is why we will pay attention mostly to per-class recall and macro-F1 score.

Dataset Identified:

For this project, the MIT-BIH Arrhythmia Database from PhysioNet will be used. It consists of 48 two-lead ECG recordings taken from 47 patients, where there are approximately 110,000 heartbeats annotated by cardiologists and sampled at 360 Hz. The beats are classified into five AAMI classes, namely N, S, V, F, and Q. To access the data, the wfdb package in Python will be used so as to retain the original ECG signal and annotations. Since around 83% of the beats are normal, class imbalance is a major issue. Beats will be segmented around R-peaks and split by record to prevent data from the same patient appearing in both training and testing.

Proposed Models for Comparison

- (i) **Random Forest:** Uses handcrafted wavelet and statistical features as a classical baseline.
- (ii) **1D CNN:** Uses raw ECG waveforms, with DE/PSO used for hyperparameter tuning.
- (iii) **BiGRU:** Uses bidirectional processing to capture patterns across the heartbeat sequence.
- (iv) **Transformer Encoder:** Uses multi-head self-attention to identify important patterns in the ECG signal.

Model-to-Member Assignment:

Member	Model owned
Tanishq Sharma	Random Forest - wavelet and statistical features
Parth Upadhyay	1D CNN with Differential Evolution/PSO
Ishaan Karmokar	BiGRU
Riddhi Puneyani	Transformer Encoder

Existing State-of-the-Art Method(s) Related to the Proposed Project:

- de Chazal et al. [1] set the inter-patient DS1/DS2 protocol (84.5% accuracy, much weaker on SVEB).
- Li et al. [2] used a Random Forest with wavelet-derived features and a novel CNN-based segment-label feature, evaluated under the inter-patient paradigm: 96% accuracy
- Kachuee et al. [3] reached 93.4% on the five AAMI classes with a residual 1D CNN;
- Salvi [4] 99.22% on the five AAMI classes using an attention-enhanced CNN-BiGRU;
- Mousavi et al. [5] 88.94% sensitivity on class S inter-patient using attention-based seq2seq with SMOTE;
- Hu et al. [6] F1 0.86 (S) and 0.96 (V) with transformer self-attention. Scores above 99% are almost always intra-patient; minority-class recall under a patient-independent split remains open.

Objectives:

- 1) Build a classifier for the five AAMI heartbeat classes, and judge it using per-class recall and macro-F1 instead of plain accuracy. Since 83% of the dataset is normal beats, accuracy alone would be misleading.
- 2) Compare all four models (Random Forest, CNN, BiGRU, Transformer) on the same train-test-split and the same metrics.
- 3) Check how well the models generalize to patients they haven't seen before, by splitting the data by record (not by individual beat) so there's no leakage between train and test.

Importance / Motivation of the Proposed Project:

Cardiovascular disease is the leading cause of death worldwide, and arrhythmias are a major contributor. A major issue here is that lethal arrhythmias can be observed for no more than a couple of seconds in hours-long recordings of ECGs, making it time-consuming and erroneous to identify dangerous events manually. Beat-level classification of events allows one to easily pinpoint those that require special attention from a cardiologist. We choose to use MIT-BIH because it is the benchmark dataset in this area, and we want our outcomes to be comparable with other published works.

Preliminary Reference List:

- [1] P. de Chazal, M. O'Dwyer, R. B. Reilly, "Automatic classification of heartbeats using ECG morphology and heartbeat interval features," *IEEE Trans. Biomed. Eng.*, vol. 51, no. 7, pp. 1196–1206, 2004.
- [2] Y. Li et al., "Heartbeat classification by random forest with a novel context feature: A segment label," *IEEE Journal of Translational Engineering in Health and Medicine* (or *IEEE Access* — check exact journal on the PMC page), 2022.
- [3] M. Kachuee, S. Fazeli, M. Sarrafzadeh, "ECG heartbeat classification: A deep transferable representation," in *Proc. IEEE ICHI*, 2018.
- [4] R. Salvi, "An explainable CNN-BiGRU model for arrhythmia classification from ECG signals," in *2024 Conference on AI, Science, Engineering, and Technology (AIxSET)*, pp. 316–319, IEEE, 2024.
- [5] S. Mousavi, F. Afghah, U. R. Acharya, "Inter- and intra-patient ECG heartbeat classification for arrhythmia detection: A sequence to sequence deep learning approach," in *Proc. IEEE ICASSP*, 2019.
- [6] S. Hu, W. Cai, T. Gao, J. Zhou, M. Wang, "Robust wave-feature adaptive heartbeat classification based on self-attention mechanism using a transformer model," *Physiol. Meas.*, vol. 42, no. 12, art. 125001, 2021.

Work Plan:

	Component / Work Element / Milestone	Expected Completion (Day/Month)
1	PhysioNet wfdb download, beat segmentation, patient split	3 Sept
2	Literature review: 8–10 papers in summary table	8 Sept
3	Random Forest baseline and RMSprop/Optuna-tuned 1D CNN implemented	14 Sept
4	BiGRU and Transformer encoder implemented	20 Sept
5	Preliminary results for all four models; interim report	25 Sept